A. 2023

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: constructive algorithms, implementation, math, number theory |
https://codeforces.com/problemset/problem/1916/A

In a sequence a, whose product was equal to 2023, k numbers were removed, leaving a sequence b of length n. Given the resulting sequence b, find any suitable sequence a and output which k elements were removed from it, or state that such a sequence could not have existed.

Notice that you are not guaranteed that such array exists.

Input

Each test consists of several test cases. The first line contains a single integer t ($1 \le t \le 100$) — the number of test cases. This is followed by a description of the test cases.

The first line of each test case contains two integers n ($1 \le n, k \le 5$) — the size of sequence b and the number of numbers removed from sequence a.

The second line contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le 2023$) — the remaining sequence. The values of b_i might not be divisors of 2023.

Output

For each test case, output "YES" if the sequence a exists, and in the following line output k non-negative integers that were removed from the sequence a. If the sequence a does not exist, output "NO" in a single line.

You can output the answer in any case (uppercase or lowercase). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive answers.

Example

Input

```
7
2 2
5 2
3 1
7 17 7
4 2
1 289 1 1
3 1
7 17 17
1 1
289
1 1
2023
1 3
1
```

Output

```
NO
NO
YES
7 1
YES
1
YES
7
YES
1
YES
7 17 17
```

Note

In third test case product is equal to $289 \cdot 7 = 2023$.

In fourth test case product is already equal to 2023.

In seventh test case product is equal to $7 \cdot 17 \cdot 17 = 2023$.